*(Please write your Exam Roll No.)* *Exam Roll No. .....*

# END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

| *Paper Code: ECC-211* | *Subject: Analog Electronics-I* |
|---|---|
| *Time: 3 Hours* | *Maximum Marks: 75* |

***Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.***

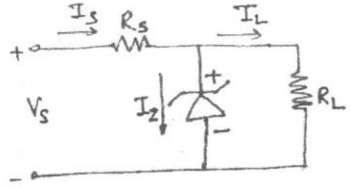
Q1 Answer **all** following questions briefly:- **(1.5x10=15)**

(a) Define the term "static resistance" and "dynamic resistance" in a *pn* junction diode.
(b) Sketch the transistor symbol for *pnp* and *npn* transistor, and then insert the conventional flow direction for each current.
(c) State the Barkhausens criterion?
(d) How does the reverse saturation current of the p-n junction diode vary with temperature?
(e) In a BJT, base current is 7.5μA and the collector current is 940μA.Find α and β for the device.
(f) What do you understand by term "Loading effect" in multistage amplifier?
(g) State what will happen to the voltage gain of the amplifier if the bypass capacitor $C_E$ is open circuited.
(h) A multistage amplifier consists of three stages .The voltage gains of the stages are 30, 50 and 80. Calculate the overall gain in dB.
(i) Draw a Darlington pair showing voltage and current component.
(j) Which type of feedback is used is used to get greater input impedance and lower output impedance in the amplifier circuits?

## UNIT-I

Q2 (a) In the circuit of the zener network below, $R_s = 20\Omega$ , $V_z = 18V, and\ R_L = 200\Omega$. If $V_S$ can vary from 20V to 30V, find: **(9)**

(i) The minimum and maximum currents in the zener diode
(ii) The minimum and maximum power dissipated in the diode.




(b) Explain the reason for the "diffusion capacitance, $C_D$" and "transition capacitance, $C_T$" in in a pn junction diode. **(6)**

Q3 (a) In a center-tap FWR, the load resistance $R_L = 1K\Omega$. Each diode has a forward –bias dynamic resistance $r_d$ of 10 Ω. The voltage across half the secondary winding is 220 sin 314t volts.Find (a)peak value of current , (b)the dc or average current (c) the rms value of current, (d) the ripple factor (e) the rectification efficiency. **(9)**

(b) Draw neatly labeled circuit diagram for Center- tap FWR and Bridge Rectifier and draw their corresponding output waveforms. **(6)**

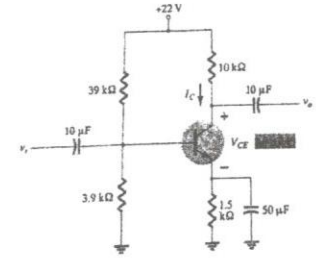
**P.T.O.**

ECC-211
P-1/2



## UNIT-II

Q4 (a) Determine the dc bias voltage $V_{CE}$ and the current $I_C$ for the voltage divider configuration of the Figure below. Given β=140. **(9)**




(b) List the sources of instability of collector current and discuss their effect on Q point. **(6)**

Q5 (a) Draw a circuit of fixed bias and self-bias, and explain qualitatively why a self - bias is better than the fixed bias circuit. Derive an expression for stability factor $s = \frac{\partial Ic}{\partial Ico}$ for both the circuits. **(9)**

(b) Write expressions of various h-parameters of a two - port network. Draw hybrid model of CE, CB and CC Transistor configuration. **(6)**

## UNIT-III

Q6 (a) Derive an expression for voltage gain of an R-C coupled amplifier at mid frequencies. What factors affect the gain of the amplifier at low frequencies and at high frequencies? **(9)**

(b) Draw labeled frequency response curve of RC coupled amplifier clearly indicating cut-off frequencies and bandwidth. **(6)**

Q7 (a) Differentiate between the constructions of enhancement type MOSFET and depletion type MOSFET. Explain the V-I characteristics of depletion type MOSFET. **(9)**

(b) Draw the structure of n-channel JFET and explain its characteristics. Write the equation for drain current, $I_D$. **(6)**

## UNIT-IV

Q8 (a) State the merits and demerits of negative feedback amplifiers. Draw the block diagram representations of the four types of negative feedback amplifiers. **(9)**

(b) Draw the circuit diagram of voltage series feedback amplifier and derive the expression for input and output impedances. **(6)**

Q9 (a) Explain the difference between voltage amplifier and power amplifier. Categorize power amplifiers by class and describe each class of power amplifier. **(9)**

(b) Explain the working of class A power amplifier. Derive an expression to show that maximum efficiency of class A amplifier is 25%. **(6)**

ECC-211